



Capital Efficiency / RWA Optimisation

Model capital released through IRB migration, collateral optimisation, securitisation, and portfolio density improvement. Built for bank CFOs, CROs, and treasury heads navigating Basel III endgame in the GCC.

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1. Executive Summary

Capital efficiency is the defining strategic lever for GCC banks navigating Basel III endgame. This accelerator provides a structured framework across four optimisation pathways to model capital release, estimate ROE uplift, and benchmark against regional peers. Each pathway includes calculation templates and implementation guidance.

2. IRB Migration & Model Sophistication

Moving from standardised to foundation or advanced IRB typically reduces credit-risk RWA by 15-35%. The magnitude depends on portfolio composition, data maturity, and regulatory approval timelines.

Migration Pathway Comparison

Approach	Typical RWA Reduction	Data Requirements	Approval Timeline
Standardised to Foundation IRB	15-20%	5+ years PD history, validated models	12-18 months
Foundation IRB to Advanced IRB	10-15% incremental	7+ years LGD/EAD history, granular data	18-24 months
Standardised to Advanced IRB	25-35%	Full internal model suite, extensive data	24-36 months

Implementation Considerations

Data Readiness: IRB approval requires demonstrably clean, complete, and validated historical data. Most GCC banks need 12-18 months of data remediation before application. **Model Governance:** SAMA expects robust model risk management frameworks aligned with SR 11-7 principles. **Regulatory Engagement:** Early and continuous dialogue with SAMA is critical. Parallel running periods typically span 6-12 months.

3. Collateral Optimisation

Recognising eligible collateral currently unrecognised under regulatory frameworks can release significant capital. The gap between economic collateral value and regulatory recognition is typically 20-40% for GCC banks.

Eligible Collateral Types

Collateral Type	Regulatory Haircut	Typical Coverage Ratio	RWA Impact
Cash & Government Securities	0-5%	Low (5-10%)	High
Listed Equities	20-40%	Medium (15-25%)	Medium

Real Estate (Commercial)	40-60%	High (25-40%)	Medium
Guarantees (Sovereign)	10-20%	Medium (10-20%)	High
Receivables	40-60%	High (30-50%)	Low-Medium

Optimisation Checklist

- Complete collateral inventory with regulatory eligibility mapping
- Valuation frequency aligned with Basel requirements
- Legal enforceability opinions current for all jurisdictions
- Haircut methodology documented and validated
- Cross-border collateral recognition assessed
- Netting agreements reviewed and updated

4. Securitisation & Balance Sheet Recycling

Significant risk transfer (SRT) via securitisation of mortgage or loan books can dramatically reduce on-balance-sheet RWA while retaining servicing economics. GCC sukuk structures offer Sharia-compliant pathways.

SRT Structure Comparison

Structure	RWA Benefit	Complexity	Market Depth (GCC)
Synthetic Securitisation	60-80% RWA reduction on retained pool	High	Emerging
True Sale Securitisation	Full derecognition possible	Medium-High	Moderate
Covered Bonds / Sukuk	Lower risk weight on retained assets	Medium	Growing
Sub-participation	Partial risk transfer	Low-Medium	Established

Key Considerations

SAMA Approval: SRT transactions require prior supervisory approval with full documentation of risk transfer.
Sharia Compliance: Structures must satisfy Sharia board requirements for asset-backed instruments.
Investor Base: GCC investor appetite for structured products is growing but still developing compared to European markets.

5. Portfolio Risk-Weight Optimisation

Systematic portfolio rebalancing, exposure netting, and guarantee restructuring to reduce average risk-weight density across the book. Typical density improvements of 1-5% are achievable.

Optimisation Levers

Exposure Netting: Consolidate bilateral exposures to reduce gross RWA. Requires enforceable netting agreements. **Guarantee Substitution:** Replace higher-risk-weight exposures with guaranteed equivalents (sovereign, multilateral). **Portfolio Rebalancing:** Shift asset mix toward lower risk-weight categories while maintaining return targets. **CRM Techniques:** Credit risk mitigation through funded and unfunded protection, credit derivatives.

Risk-Weight Density Template

Asset Class	Current Density	Target Density	Actions
Corporate (Investment Grade)	75-100%	50-75%	IRB migration, collateral recognition
Retail Mortgages	35-75%	20-35%	Securitisation, LTV improvement
SME	75-100%	50-75%	Guarantee programmes, data improvement
Sovereign / PSE	0-20%	0-10%	Rating upgrade, ECAI alignment
Interbank	20-50%	20-35%	Netting, collateral posting

6. Capital Release Estimation Framework

Calculation Template

Use this template to estimate total capital release across all four pathways:

Pathway	Input	Formula	Estimated Release
IRB Migration	Current RWA, migration factor	$RWA \times (target\ factor - current\ factor)$	[Calculate]
Collateral	Unrecognised collateral, eligible ratio, haircut	$Unrec\ Collateral \times eligible\ ratio \times haircut$	[Calculate]
Securitisation	Eligible book, current RWA, target RW	$Book \times (current\ RW - target\ RW)$	[Calculate]
Portfolio Density	Total RWA, density improvement	$RWA \times density\ improvement$	[Calculate]
Total			[Sum]

ROE Impact Estimation

Rule of thumb: Each SAR 1B of released capital translates to approximately 0.8% ROE uplift (assuming 12.5% CAR and current net income margins). **AT1 Sukuk equivalent:** Released capital reduces the need for

expensive AT1 instruments. Each SAR 1B of capital release saves approximately SAR 150M in AT1 sukuk equivalent.

7. Scoreboard & GCC Benchmarks

GCC Banking Capital Release Benchmarks

Scenario	Capital Released (SAR B)	ROE Impact	Typical Profile
Conservative	0.5	+0.4%	Single pathway, standardised approach
Base Case	2.5	+2.0%	Two pathways, foundation IRB
Aggressive	5.3	+4.2%	All four pathways, advanced IRB

Note: These benchmarks are based on analysis of GCC Tier 1 banks with total assets of SAR 200-500B. Actual results will vary based on portfolio composition, regulatory approval timelines, and market conditions.

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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